SUPPLEMENTARY MATERIAL FOR "KNOWING THE STORE: WHAT A MEMORY BACKEND MUST WRITE DOWN BEFORE AN AGENT CAN READ IT"

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This supplement holds the records the paper's appendices point to: the instrument ledger, the deviation index, the prediction texts written before scoring, the per-provider run disclosures, the map from the artifact's internal names to the paper's terms, and the pilot table, the caching-economics paragraph and the neutral-slice outcome sentence moved here from the paper. Each block is printed as it stood in the paper, with its label, so that a pointer in the paper resolves here; every number in it also prints in a table of the paper.

S.1 THE MONITORING-SCOPED INSTRUMENT LEDGER

The twelve entries of Appendix D of the paper, and the account of the four failures of the same kind in the manuscript's own verification battery.

1. **What went wrong.** The expiry route that ignored query intent looked mechanical and was treated as one. It reached 20% precision. **What we changed.** PN is defined conditionally on intent, which is the definition this paper uses.
2. **What went wrong.** The half of PN labeling that looked mechanical reached 69–86% precision under spot check. **What we changed.** Every candidate receives human verification, and no part of the label is taken as mechanical.
3. **What went wrong.** The pilot's monitor prompt asserted lifecycle structure that two of the three backends do not expose, so it stated something false to them. **What we changed.** The prompt was rebuilt before the main study, which is why pilot and main-study values are not comparable.
4. **What went wrong.** Lexical presence recall was 0.697, low enough that every ABSENT label rested on a weak check. **What we changed.** A two-channel replacement reached 0.995, so its miss rate on present targets is $\varepsilon = 0.005$. That is a conditional rate and not the error rate of an absence claim; the posterior, which also needs the base rate, is given in Section 4.
5. **What went wrong.** Wall-clock ingestion time would have converted a measurement-step error into a lifecycle claim. **What we changed.** A proper recency time was added.
6. **What went wrong.** The probe screen fired on every content word. **What we changed.** It was replaced by a corpus-measured filter.
7. **What went wrong.** Only $3/300$ GPT-5.6 Sol scores and $1/300$ GPT-5.6 Luna scores reproduced across five draws. **What we changed.** The $k = 5$ frontier protocol followed.
8. **What went wrong.** A frontier conclusion had already been drafted when that redraw ran. **What we changed.** The preregistered redraw reversed it before distribution.
9. **What went wrong.** The frontier provider/API's "tokens used" field reports a turn total rather than output telemetry. **What we changed.** Exact frontier economics are not reported.
10. **What went wrong.** An import-shadowing bug wrote this study's usage records into another study's cache. **What we changed.** It is disclosed here rather than repaired retrospectively.
11. **What went wrong.** A label-supervised question-only probe exposed generator-specific template signal. **What we changed.** The powered matched set is reported as a second population rather than as evidence controlled for the question signal.
12. **What went wrong.** A term-frequency summary-reader baseline (NB-3) read the question rather than the store; Appendix E.8 holds the record. **What we changed.** NB-3 was disqualified as a summary reader and is retained as a measurement-step failure record.

Preventive. **What went wrong.** The generic frontier-family model string routed server-side to the top tier. **What we changed.** Exact model-identifier verification caught the mismatch before any call.

Across the ledger, the shared failure is a semantic assumption hidden inside a seemingly mechanical operation. The same failure appeared four times in this manuscript's own verification battery. That battery sits outside the monitoring scope this ledger covers, and it is recorded here because the pattern is this ledger's subject. A numeric re-scan defaulted to the frozen manuscript, so it audited a file nobody edits. A published figure passed that scan by matching an unrelated interval endpoint of equal value. An appendix exemption keyed by section name authorized one edit and then passed every later edit to the same section. Those three reported success for a reason unrelated to the question asked. The fourth is the cleanest case: an exemption table named the same section twice, and the language runtime discarded the earlier entry at parse time, so that rule had never been in force and no run could have exposed it.

S.2 Deviation index

Every deviation the paper names, with its ledger entry and the appendix that discusses it (Appendix F of the paper).

Deviation index.
Intent conditioning: E0 D16; Appendix D, item 1.
PN verification: registered Phase-0 resolution; Appendix D, item 2.
Prompt structure: E1 D2; Appendix D, item 3.
Presence control: E1 D5–D6; Appendix D, item 4.
Recency time: E1 D13; Appendix D, item 5.
Probe screen: E1 D15; Appendix D, item 6.
Model nondeterminism: E1 D19 and D21; Appendix D, item 7.
Redraw conclusion: E1 D21; Appendix D, item 8.
Provider/API telemetry: E1 D24; Appendix D, item 9.
Import shadowing: E1 D25; Appendix D, item 10.
Per-provider run disclosures: D86–D91; Appendix B.
Cross-vendor failure-text defect: D93; Appendix B.
Question-authorship confound: D94; Appendix C.
Question-reading baseline (NB-3): E1-R D-E1R-2; Appendix D, item 12.
Withdrawn coverage prediction (NB-6): E1-R D-E1R-7; Appendix E.8.
Obligation-margin correction after NB-6: E1-R D-E1R-8; Appendix E.8.
Own-name census key collision: E1-R D-E1R-10; Appendix E.8.
Retraction quota on the measured eligibility rate: CF-2 D-CF2-1; Appendix C.2.
Replacement in walk order after an authoring or reading failure: CF-2 D-CF2-2; Appendix C.2.
One re-authoring of a record that fails the author's read: CF-2 D-CF2-3; Appendix C.2.
No further queue extension; the supersession quota drops by the failures: CF-2 D-CF2-4; Appendix C.2.
Drop rule evaluated on the 13th, once: CF-2 D-CF2-5; Appendix C.2.
Truthful metadata does not flag 22 of the 104 forbidding records: DS D-DS-1; Appendix E.10.
Fetch-set zero count transcribed from the unflagged count: DS D-DS-2; Appendix E.10.
Two bars set under the premise corrected at L2; one prediction unfulfillable by its own gate: DS D-DS-3; Appendix E.10.
No deviation on the mechanism tests (RESP); Appendix E.10.
Zero-change wording of the DS-P4 failed form: DS D-DS-4; Appendix E.10.
Exact model identifier: E1.7 preventive check; Appendix D, Preventive.

S.3 Texts written before scoring, printed with their numbers

The eleven prediction texts of the mechanism tests and the downstream experiment (Appendix E.10 of the paper), each with its outcome; the placeholders were filled from the ledgers at print time and no word was changed.

Texts written before scoring, printed with their numbers. Own-name ablation, supersession (L1-P1, Table 40; held). Removing the target's own name and its nearest neighbours from the superseded-names section costs the smallest model 0.3262 of its supersession discrimination (0.8572 to 0.5310, interval [-0.3991,-0.2548]; persona-clustered [-0.4327,-0.2167]): the obligation is read off the name, as the manuscript says.

Own-name ablation, the two counted states (L1-P2, Table 40; failed). On retraction-derived items, which the summary only counts and never names, the ablation moves the smallest model's discrimination by -0.1404 with interval [-0.2569,-0.0508] (persona-clustered [-0.3305,-0.0028]), outside the tolerance. The change cannot come from a name the positives carried; it comes through the negatives whose names were removed, or through the model's response to a section that now reads "(none)" against non-zero counts. The per-state reading of Table 20 is not clean of the ablation on this state. Eleven of the 26 retraction-derived obligations carry their own name in the section the ablation edits; the row prints with that census beside it.

Retrieved view against the summary (L2-P1, Table 40; failed). A monitor shown the five most similar entries with their lifecycle metadata, and no summary, scores 0.6851 on obligation against 0.7339 under the truthful summary, a paired difference of -0.0488 with interval [-0.1274,+0.0285] (persona-clustered [-0.1454,+0.0389]) that does not fall below the summary by more than the tolerance. On this population a five-item view carries the obligation signal the summary carries; the argument that a bounded retrieved view cannot settle the judgment is not supported at k = 5 here.

Summary with the retrieved view (L2-P2, Table 40; failed). Adding the five retrieved entries to the truthful summary raises the smallest model's obligation discrimination from 0.7339 to 0.8303, a paired difference of $+0.0964$ with interval [+0.0415,+0.1554] (persona-clustered [+0.0227,+0.1635]) above the tolerance. Retrieved items carry obligation signal the summary does not: the summary names superseded content and only counts the other two states, and a retrieved entry flagged retracted or expired is a name the summary never prints.

Reach of the forbidding record (L2-P3, Table 40; failed). Over every entry the store holds, a five-item similarity view contains the record that forbids use for 54 of the 104 obligations, a reach rate of 0.5192. The record does not rank below the cutoff on this population for a retriever that does not mask inactive entries; the 5% figure the manuscript cites is FR-Bank's AV7 retrieval-pass rate under its lifecycle mask, a different quantity, and the premise sentence is to be restated as a mask effect.

The rubric gate (L3-P1, Table 40; failed). With the detection step removed and the probability elicited directly, the smallest model's blind obligation score still takes 2 distinct values across the 661 questions (604 at the modal value). The constant was not the gate: the reviewer's explanation is not supported, and the smallest model's blind obligation judgment is flat under either rubric.

Downstream, the oracle ceiling (DS-P1, Table 40; failed). Fetching the store's lifecycle records for the obligation's own attribute, on gold labels, puts the forbidding record in the view for 99 of the 104 obligations against 54 for the five-entry similarity view alone, a difference of $+0.4327$ (interval [+0.3365,+0.5288]; persona-clustered [+0.2815,+0.5680]) that does not exceed 0.50. Either the retrieved view already carries the record for most obligations, or the store's metadata does not flag it: the truthful join marks the forbidding record non-active for 82 of 104 (8 of 26 retraction-derived, 11 of 15 expiry-derived), and no fetch by lifecycle state reaches the other 22. Acting on a perfect obligation judgment reaches 0.9519 on this store, not one.

Downstream, retention of the oracle gain (DS-P2, Table 40; failed). On the 63 supersession-derived obligations, where every forbidding record is flagged and named, gating the fetch on the smallest model's recorded obligation score at its base rate puts the record in view for 0.8413, against 0.3492 under retrieval alone and 1.0000 under gold labels, retaining 0.7561 of the oracle gain with interval [0.6097,0.8718] (persona-clustered [0.5625,0.9394]) that does not exclude 0.60 from below. The gate fires on 31 of the 41 supersession-derived obligations the retriever misses; the summary supports the judgment, and the judgment at this threshold does not carry it into the view. The item-level and the persona-clustered intervals disagree on this prediction; the weaker reading governs.

Downstream, the lookup as the rule (DS-P3, Table 40; held). The superseded-names lookup, at the same fetch budget, puts the record in view within tolerance of the model's gate (0.8413 against

0.8413 on the 63, +0.0000, interval [-0.0794,+0.0794]; persona-clustered [-0.0938,+0.1035]): on the state the summary names, the gate can be a lookup.

Downstream, naming the counted states (DS-P4, Table 40; failed). Naming the retracted and expired attributes in the summary makes their records fetchable, and the gate elicited under the counting summary reaches them for 0.7805 against 0.7805 under the superseded-only names, a difference of +0.0000 (interval [+0.0000,+0.0000]; persona-clustered [+0.0000,+0.0000]) that does not exceed 0.20 on the 41 retraction- and expiry-derived obligations. Of the 41, 19 carry a flagged record; naming makes 17 of them fetchable that the superseded-only names do not, and the gate fires on 0 of those. The bound on these states is the judgment, not the naming.

Downstream, the coverage rule (DS-P5, Table 40; held). The gate does not cost coverage: on the 174 items whose licensed record the store holds, coverage under the gated policy is within tolerance of retrieval alone (0.3448 against 0.3736, −0.0287, interval [-0.0519,-0.0057]; persona-clustered [-0.0680,-0.0050]).

S.4 Per-provider disclosures and the failure-text correction

The run disclosures for the three new providers of the cross-vendor replication (Appendix B of the paper) and the correction of one frozen failure explanation.

Per-provider disclosures (D86–D91; Appendix F). In order: Haiku accepted the frozen temperature, Opus omitted an unsupported temperature, and K3 accepted only its provider-required value. K3 required a larger output ceiling to emit parseable content. A system-role assembly difference between the new providers and the OpenAI frontier path makes the indexed-coverage band item a band comparison rather than a matched contrast. Provider-specific tokenization is recorded for economics only. K3 is pinned to Moonshot direct rather than a routing mixture.

The five-draw census (R107; `reviews/NUMBERS_R107.md`). On the recorded five-draw lanes, Haiku 4.5's obligation score is identical across the five draws on 85% of the item-arm groups of the 661 (1,123 of 1,322; median range 0), and its reply text differs on 42% of them while the number stays. Opus 5's and GPT-5.6 Luna's scores differ across the five draws on 97% and 96% of the groups (median range 0.07 and 0.05), because Opus 5's provider rejects the temperature parameter and the reasoning lanes omit it, so averaging the five draws is what makes those rows stable. The census script and its output accompany the artifact. Under that always-reasoning configuration the preregistration pilot measured a mean of 1,027 reasoning tokens per call. The measured Opus output-ceiling correction was made before the full run.

Failure-text correction (D93; Appendix F). The frozen failure explanation for the universal-blind item (XV-1; Appendix F) assumed that any miss would be above chance and that the indexed separation would fail with it. Haiku instead missed below chance, while every new indexed-minus-blind NOK interval excluded zero. The registered gate and its failed verdict remain unchanged; the attached explanation, rather than the gate, was defective.

S.5 Internal names in the artifact

The third glossary block of Appendix E.1 of the paper: every internal name a reader meets in the registrations, ledgers and row files, and the term the paper prints for it.

Internal names in the artifact, and the term this paper prints for each.	
E0 / E1	the pilot / the main study
lineage	one model family and version
lane	one model scored on one question set
cell	a fixed question set used for one comparison
prior	what the question alone gives away, measured by the blind score
arm	one experimental condition
provenance	which summary a row used, TRUE or BELIEVED
committed	recorded and frozen before later analysis
registered	preregistered, with the prediction fixed before data collection
NB-1 . . . NB-6	the six mechanical readers of the same summary
NB-6	the superseded-names lookup, scoring the query against the flat superseded-names list
cross-model cell	cross-model set (84): the matched 84-question set
authorship-matched slice	authorship-matched set (36): the 36-question set rewritten in one voice
counterfactual cell	store-counterfactual set (444): 148 queries asked of three versions of one store; its first form had 30
count-destruction probe	the test that randomizes every count in the summary
SWAP control	scoring the monitor against another persona's summary
D-numbers	entries in the deviation ledgers of Appendix F
index, metadata index, store description	the summary (Definition 2; "lifecycle index" is its formal name)
self-description	what a backend writes about itself; the summary, when it is the object read
question-only arm, blind arm	the blind condition
indexed arm	the indexed condition
delta, paired delta, indexed-minus-blind	the gain
matcher, NB-6, name lookup	the superseded-names lookup
NB-1 . . . NB-5	the other mechanical readers of the summary (Appendix E.8)
mechanical reader	any of the six non-model readers of the summary
lineage, lane	the model, named in full (GPT-4.1 mini, GPT-5.6 Luna, GPT-5.6 Sol, Opus 5, Haiku 4.5, Kimi K3)
the smallest model, the largest model	GPT-4.1 mini; GPT-5.6 Sol on the full populations and the cross-model set, and Opus 5 on the counterfactual set, where GPT-5.6 Sol was not run
slice, population	question set (Section 4); "full populations" is a set name
committed, registered, frozen	recorded before scoring; preregistered; the text written before scoring
witness, PN record, the record that forbids use	the forbidding record
inside the band, inside the tolerance band	within tolerance (± 0.05)
mixed, by the weaker convention	the weaker reading governs
no verdict, carries no prediction, reports, anchors no gain	descriptive (no prediction)

S.6 STRUCK CLAUSES

The clause of a frozen failure text that was struck as defective, kept here as the record that Appendix D.2 of the paper points to; the paragraph is printed as it stood in the paper.

The registered failure text continued that the gain measured on the larger population is therefore not free of what the question alone gives away. That clause is struck as defective and is not reported (deviation ledger, Appendix F). At this size the set can neither establish the aggregate effect nor establish contamination: the point estimate is positive, the two interval conventions disagree, and the largest model tested, on which no prediction was registered, reaches 0.7122 under the same summary with an item-level interval that excludes chance.

S.7 PILOT ESTIMANDS

The pilot preceded the main study with a different prompt, so its values support the design lesson and cannot be compared numerically with the main study (Table 1). The raw and blind-deduplicated coverage estimands must not be mixed for the same reason.

Table 1: The pilot table records its FOK, NOK, and calibration estimands. It supports the design lesson but cannot be compared numerically with the main study because the prompts differ.

Result	n	Blind / estimate	Indexed / target	Δ	95% CI
FOK, raw	210 (141/69)	0.5087	0.8176	$+0.3088$	$[+0.2051, +0.4090]$
FOK, blind-deduplicated	not recorded	0.5429	same indexed condition	$+0.2747$	not recorded
NOK	not recorded	0.4909	0.5320	$+0.0411$	$[-0.0569, +0.1255]$
Non-forced calibration	not recorded	$\sim$0.26 mean	67% base rate	ECE 0.4155	not recorded

S.8 CACHING ECONOMICS

Monitoring runs before every query, so its cost turns on prompt caching. The list-price projection and the billed micro-run disagree about which monitor is cheaper. With caching the indexed monitor is projected at $207 per million calls against $234 blind, while as billed it costs $360 against $233. The difference arises because cached input bills at 25% of list and 64.3% of indexed prompt tokens hit the block-aligned cache. The deployment Section 10 recommends makes no model call for obligations, so both orderings concern the model monitor only.

S.9 THE NEUTRAL-SLICE OUTCOME SENTENCE

The registered outcome sentence of the neutral-slice prediction, as the paper's Limitations section printed it, reads: The indexed NOK gain excludes zero for Opus 5 and includes zero for GPT-4.1 mini, which we record as a failed prediction.

S.10 PROPOSITION 1 AND ITS PROOF

Remark 1 in Section 3 of the paper is the informal form of the proposition below, which the preregistered texts call Proposition 1; the paper prints the remark and its justification, and the formal statement and proof are here (R108, the author's instruction of 2026-09-14). The construction builds two stores that a top-k retriever cannot tell apart, which is what makes the obligation undetectable from a bounded retrieved view.

S.10.1 FORMAL STATEMENT AND PROOF

Proposition 1 (Monitoring non-identifiability)**.** *Fix k and a per-item score σ whose retriever masks non-active records from its ranking, as lifecycle-tracking and overwriting stores do; under that mask a lifecycle record that changes intent-conditional coverage ranks below k common items (on the inherited store, retraction records pass at 5%). Then there exist S, S' with feature-identical retrieved sets $R_k(q, S) = R_k(q, S')$ but $\mathrm{Cov}(q, S) =$ PRESENT and $\mathrm{Cov}(q, S') =$ PN; under the uniform distribution over $\{S, S'\}$, every decision rule over the monitor's output errs with probability at least $1/2$.*

Fix k, the score σ, and a below-cutoff witness from Proposition 1; let q ask for the user's current weekend plan. Let e_g be an active item ("hiking trip Saturday") with slot key `weekend_plan` and $\sigma(e_g, q) = 0.95$, and let $D = \{d_1, \ldots, d_{k-1}\}$ be lifecycle-inert filler items satisfying $0.85 < \sigma(d_j, q) < 0.90$. Set $S = \{e_g\} \cup D \cup F$ and $S' = \{e_g\} \cup D \cup F \cup \{e_r\}$, where every item in the common pool F has similarity below 0.80 and e_r is a retraction of the same slot with $\sigma(e_r, q) = 0.70$. The retraction lies below the fixed ranker's cutoff, so that ranker retrieves the same feature-identical top-k set from both stores. Under current intent, $\mathrm{Cov}(q, S) =$ PRESENT and $\mathrm{Cov}(q, S') =$ PN. The monitor's input, and hence output, is identical on the two stores, while exactly one required verdict is wrong; under their uniform mixture, its error is at least $1/2$. □